Multiple Choice Questions

1. Conductors for high voltage transmission lines are suspended from towers

(A) to reduce clearance from the ground
(B) to increase clearance from ground
(C) to reduce wind and snow loads
(D) to take care of extension in length during summer

B
4. The fact that a conductor carries more current on the surface as compared to the core, is known as

(A) skin effect
(B) corona
(C) permeability
(D) unsymmetrical fault.

A
7. Skin effect depends on

(A) Size of the conductor
(B) Frequency of the current
(C) resistivity of the conductor material
(D) all of the above.

D

9. For constant voltage transmission the voltage drop is compensated by installing

(A) Synchronous motors
(B) Capacitors
(C) inductors
(D) all of the above

B
11. The effect of the corona is

(A) Power loss
(B) Radio interference
(C) Ozone formation
(D) all of the above.

D
13. Between two supports, due to sagging the conductor takes the form of

(A) Catenary
(B) Triangle
(C) ellipse
(D) semi-circle.

A
14. The effect of ice deposition on a conductor is

(A) Increased skin effect
(B) reduced corona losses
(C) increased weight
(D) reduced sag.

C
15. The effect of wind pressure is more predominant on

(A) Transmission lines
(B) Neutral wires
(C) insulator
(D) supporting towers.

A
17. For improving life, steel poles are galvanized.
Galvanizing is the process of applying a layer of

(A) Paint
(B) Varnish
(C) tar coal
(D) zinc.

D
18. The disadvantage of transmission lines as compared to cables is

(A) Exposure to lightning
(B) Exposure to atmospheric hazards like smoke, ice, etc.
(C) Inductive interference between power and communication circuits
(D) all of the above.
D
19. ACSR conductor implies

(A) All conductors surface treated and realigned
(B) Aluminum conductor steel reinforced
(C) Anode current sinusoidally run
(D) Anodized Core Smooth Run.

B
20. During a storm the live conductor of the public electric supply breaks down and touches the earth. The consequences will be

(A) The supply voltage will drop
(B) The supply voltage will increase
(C) current will flow to the earth
(D) no current will flow in the conductor.
C

21. In the transmission system a feeder feeds power to
(A) Service mains
(B) Generating stations
(C) distributors
(D) all of the above.

Feeder→Distributor→Service mains→Consumer

23. The chances of corona are maximum during
(A) Summer heat
(B) Winter
(C) Dry weather
(D) humid weather.

D

12. By the use of which of the following power factor can be improved?

(a) Phase advancers
(b) Synchronous compensators
(c) Static capacitors
(d) Any of the above

d

13. Which of the following is the disadvantage due to low power factor?

(a) Poor voltage regulation
(b) Increased transmission losses
(c) High cost of equipment for a given load
(d) All of the above

d

14. Base load plants usually have ..... capital cost, ... operating cost and ..... load factor.

(a) high, high, high
(b) high, low, high
(c) low, low, low
(d) low, high, low

b

15. For a consumer the most economical power factor is generally

(a) 0.5 lagging
(b) 0.5 leading
(c) 0.95 lagging
(d) 0.95 leading

c

17. Most efficient plants are normally used as

(a) peak load plants
(b) base load plants
(c) either (a) or (b)
(d) none of the above

B (peak load plants : verimden daha çok, hızlı devreye girme özelliği önemlidir. Ama base load sürekli çalıştırıılarak daha fazla enerji üretmede daha verimlidir.)

17. For the same maximum demand, if load factor is decreased, the cost generation will

(a) remain unchanged
(b) decrease
(c) increase

C (Load factor azalınca aynı kurulu kapasiteyle daha az enerji üretilir)

18. In a load curve the highest point represents

(a) peak demand
(b) average demand
(c) diversified demand
(d) none of the above

a

19. The efficiency of a plant is of least concern when it is selected as

(a) peak load plant
(b) casual run plant
(c) either (a) or (b)
(d) base load plant
a

20. The purpose of providing an iron core in a transformer is to

A. Provide support to windings
B. Reduce hysteresis loss
C. Decrease the reluctance of the magnetic path
D. Reduce eddy current losses
c

21. Load shedding is done to (üretilen enerjnin tüketileni karşılayamadığı durumlarda bazı bölgelerde bilinçli elektrik kesilmesi)

(a) improve power factor
(b) run the equipment efficiently
(c) repair the machine
(d) reduce peak demand

d

22. Which of the following is the essential requirement of peak load plant?

(a) It should run at high speed
(b) It should produce high voltage
(c) It should be small in size
(d) It should be capable of starting quickly

d

23. Which of the following is not an operating cost?

(a) Maintenance cost
(b) Fuel cost
(c) Salaries of high officials
(d) Salaries of operating staff
c

25. A nuclear power plant is invariably used as a

(a) peak load plant
(b) base load plant
(c) stand-by plant
(d) spinning reserve plant
(e) any of the above
b

26. Transmission lines are transposed to

(a) reduce copper loss
(b) reduce skin effect
(c) prevent interference with neighboring telephone lines
(d) prevent short circuits between any two lines
c

27. Which of the following are the advantages of interconnected operation of power systems?

1. Less reserve capacity requirement
2. More reliability
3. High power fact
4. Reduction in short-circuit level

Select the correct answer using the codes given below:

(a) 1 and 2
(b) 2 and 3
(c) 3 and 4
(d) 1 and 4

a

[1] Why is the efficiency of transformers more than that of other rotating machines?

There are no moving parts in transformer hence the friction and mechanical losses are absent in transformer. Hence efficiency of the transformer is more than of other rotating machines.
(Transformatörde hareketli parça bulunmadığı için:

- Sürtünme kaybı,
- Rüzgâr/sürtünme mekanik kayıpları

yoktur. Temel kayıplar bakır ve çekirdek kayıplarıdır.)

[2] What material is used in the transformer core?

High grade silicon steel laminations are used for the construction of the core.

[3] What is an ideal transformer?

A transformer is said to be ideal if it satisfies following properties:

- It has no losses.
- Its windings have zero resistance.
- Leakage flux is zero, i.e., 100% flux produced by primary links with the secondary.
- Permeability of core is so high that negligible current is required to establish the flux in it.

- Kayıp yoktur.
- Sargı dirençleri sıfırdır.
- Leakage flux yoktur.
- Primer akısının tamamı sekonderi bağlar.
- Çekirdek geçirgenliği sonsuz kabul edilir.
- Magnetizing current ihmal edilecek kadar küçüktür.

Bu nedenle ideal transformatörde:

$\eta=100\%$

[6] Power transformers are generally designed to have maximum efficiency around

A. No load
B. Half load
C. Near full load
D. 10% overload

c

[7] The transformer laminations are insulated from each other by

A. Mica strip
B. Thin coat of varnish
C. Paper
D. Any of the above

b

[8] A transformer core is laminated to

A. Reduce hysteresis loss
B. Reduce eddy current losses
C. Reduce copper losses
D. Reduce all above losses

B (girdap akım kayıplarını azaltır)

[9] A transformer oil must be free from

A. Sludge - çamur
B. Odour - koku
C. Gases -gaz
D. Moisture- nem

d